

Short Term Internship on AI vs Human Text Detector



A Short-Term Internship report submitted in partial fulfilment for the
award of degree in

**BACHELOR OF TECHNOLOGY
IN
CSE (Data Science)**

By

**CHENNA REDDY GARI VISHNU VARDHAN REDDY
229X1A3231**

Under the esteemed guidance of

**Sri S. Md. Nadeem
Assistant Professor
Department of CSM**

Department of CSE (AI & ML)

**G. PULLA REDDY ENGINEERING COLLEGE(Autonomous): KURNOOL
(Affiliated to JNTUA, ANANTAPURAMU)**

2025-2026

Department of CSE (AI & ML)

G. PULLA REDDY ENGINEERING COLLEGE(Autonomous): KURNOOL

(Affiliated to JNTUA, ANANTAPURAMU)



CERTIFICATE

This is to certify that the Short-Term Internship on AI vs Human Text Detector, is a Bonafide record of work carried out by CHENNA REDDY GARI VISHNU VARDHAN REDDY (229X1A3231) under my guidance and supervision in partial fulfillment of the requirements for the award of degree of BACHELOR OF TECHNOLOGY IN CSE (Data Science) from G. Pulla Reddy Engineering College, Kurnool.

Sri S. Md. Nadeem

Assistant Professor,
Department of CSM,
GPREC,
KURNOOL.

Dr. R. Praveen Sam

Professor & HOD,
Department of CSM,
GPREC,
KURNOOL.

Program Book

Short Term internship

G. PULLA REDDY ENGINEERING COLLEGE(Autonomous): KURNOOL

(Affiliated to JNTUA, ANANTAPURAMU)

2025-2026

Program Book for Short-Term Internship

Name of the Student : CHENNA REDDY GARI VISHNU VARDHAN REDDY

Name of the College : G. Pulla Reddy Engineering College

Registration Number : 229X1A3231

Period of Internship : 6 Weeks From: 05/05/2025 To: 20/06/2025

**Name & Address of
the Intern Organization** : SkillDzire, Hyderabad, Telangana, India.

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education <https://apsche.ap.gov.in>

1. It is mandatory for all the students to complete 2 months (180 hours) of short-term internship either physically or virtually.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements of transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the persons working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your college has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned with a Faculty Guide from your College. He / She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
 - Data and Information you are expected to collect about the organization and/or industry.
 - Job Skills you are expected to acquire.
 - Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.
13. Never be hesitant to ask questions to make sure you fully understand what you need to do your work and to contribute to the organization.

- 14.** Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
- 15.** At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
- 16.** There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
- 17.** Do not meddle with the instruments/equipment you work with.
- 18.** Ensure that you do not cause any disturbance to the regular activities of the intern organization.
- 19.** Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
- 20.** You should understand that during internship programme, you are the ambassador of your college, and your behavior during the internship programme is of utmost importance.
- 21.** If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
- 22.** Do not forget to keep up your family pride and prestige of your College.

An Internship Report
On
AI vs Human Text Detector

Submitted in accordance with the requirement for the degree of B. Tech in CSD

Name of the College : G. Pulla Reddy Engineering College.
Department : CSE (AI&ML)
Name of the Faculty Guide : Sri S. Md. Nadeem
Duration of the Internship : 6 Weeks
From : 05/05/2025
To : 20/06/2025
Name of the Student : CHENNA REDDY GARI VISHNU VARDHAN REDDY
Program of Study : CSD
Year of Study : IV B.Tech (2025-2026)
Register Number : 229X1A3231
Date of Submission : 15/07/2026

Student's Declaration

I, CHENNA REDDY GARI VISHNU VARDHAN REDDY, a student of B. Tech Program, Reg. No. 229X1A3231 of the Department of CSE (AI&ML) in G. Pulla Reddy Engineering College do hereby declare that I have completed the mandatory Short-term Summer internship in AI vs Human Text Detector from May 5th, 2025 to June 20th, 2025 in SkillDzire under the Faculty Guideship of Sri S. Md. Nadeem, Assistant Professor, Department of CSE (AI&ML) in G. PULLA REDDY ENGINEERING COLLEGE.

(Signature and Date)

Endorsements:

Faculty Guide : Sri. S. Md. Nadeem, Assistant Professor
Head of the Department : Dr. R. Praveen Sam, Professor & HOD
Principal : Dr. B. Sreenivasa Reddy.

SkillDzire

CIN Number : U74999TG2020PTC144924

Date - 12-05-2025

Dear Chenna Reddy Gari Vishnu Vardhan Reddy,

Congratulations on your successful enrollment in the AI vs Human Text Detector Internship Training Program at SkillDzire Technologies Pvt. Ltd.

We are pleased to confirm your registration under Student Registered No. 229X1A3231 from the CSE - Data Science Department of G Pulla Reddy Engineering College.

This internship program is designed to enhance your skills and knowledge in AI, Machine Learning, and Software Engineering. We are thrilled to have you on board and are committed to ensuring a productive and enriching learning journey for you.

We look forward to supporting your growth and success during this internship.

Your Sincerely,

Srikanth Muppala (Founder)
Skilldzire Technologies Pvt Ltd.

SkillDzire Technologies

CERTIFICATE OF INTERNSHIP

This is to Certify that Mr./Ms
CHENNA REDDY GARI VISHNU VARDHAN REDDY

Enrolled in the CSE- Data Science - 229X1A3231
From College G Pulla Reddy Engineering College
of university JNTUA, Anantapur

has Successfully Completed short-term Internship programme titled
AI vs Human Text Detector

Under SkillDzire from 05-May-2025 to 20-June-2025 Organized By SkillDzire

Certificate ID: SDST-25-14019

Issued On: 20-June-2025

Authorized Signature

ACKNOWLEDGEMENTS

I am thankful to my Guide Sri. S. Md. Nadeem Sir, Assistant Professor, Department of CSE (AI & ML), on his valuable guidance and encouragement. His helping attitude and suggestions have helped me in the successful completion of the project report.

I would like to express my gratefulness and sincere thanks to my HOD Dr. R. Praveen Sam Sir, Professor & HOD for his kind help and encouragement during the course of our study and in the successful completion of the project report.

I express my thanks to my beloved Principal Dr B. Sreenivas Reddy Sir, of G. Pulla Reddy Engineering College. I would like to thank my parents and friends, who have the greatest contributions in all achievements, for the great care and blessings in making as successful in all my endeavors.

I would like to express my gratitude to all my teaching faculty and non-teaching staff for their kind help, encouragement for the great care and blessings in making as successful in all my endeavors.

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CHAPTER 1: EXECUTIVE SUMMARY

Activities Done:

1. Exploration of Java 17 and Spring Boot microservices patterns, including service registration and central routing gateways.
2. Developing a microservices architecture comprising user-service, detection-service, api-gateway, and frontend-service.
3. Integrating MySQL relational database with Spring Data JPA for secure user management and storage.
4. Implementing a Flask-based Python NLP service using PyTorch and Hugging Face pre-trained GPT-2 models.
5. Calculating text perplexity dynamically based on token cross entropy loss and NLTK-based text burstiness metrics.
6. Calibrating classification thresholds (Perplexity: 25, Burstiness: 0.20) to identify human vs machine-generated text.
7. Designing interactive HTML5/Bootstrap UIs rendered with Thymeleaf, utilizing AJAX and Plotly.js charts for visualization.

Learning Objectives:

1. To acquire technical and domain-specific skills in Java Spring Boot and Spring Cloud microservices.
2. To understand real-world application of Natural Language Processing (NLP) models.
3. To build secure and scalable relational database connections with Hibernate ORM.
4. To understand service discovery and gateway request routing in enterprise backend architectures.

Learning Outcomes:

1. Proficiency in developing, packaging, and running multi-module Maven microservice architectures.
2. Capability to deploy pre-trained NLP transformer models for text categorization.
3. Experience in front-end template rendering with Thymeleaf and dynamic data plotting with Plotly.
4. Understanding of Spring Cloud Gateway routing configurations and Eureka Service Registries.
5. Experience in debugging logic bugs (logit token shifting) and networking errors (CORS / relative gateway ports).

CHAPTER 2: OVERVIEW OF THE ORGANIZATION

SkillDzire is one of India's leading real-time learning platforms, dedicated to offering job-oriented courses, internships, and training programs to students and professionals. With a strong vision to bridge the industry skill gap, SkillDzire focuses on providing practical, project-based learning opportunities aligned with current market demands.

The organization began with a small team of six employees and has now grown to a workforce of over forty professionals. Courses and internships are curated by more than twenty industry experts, ensuring high-quality, industry-relevant content.

Key Features of SkillDzire:

- **Learner-Centric Approach:** Courses are interactive, engaging, and aligned with modern industry practices.
- **Hands-On Learning:** Programs include real datasets, live projects, and assessments to track progress.
- **Comprehensive Resources:** Participants have access to structured course materials, practical exercises, and mentor guidance.
- **Diverse Programs:** The organization offers multiple job-oriented courses across domains like Data Science, Web Development, AI, and Machine Learning.

Vision: To empower students and professionals with the skills, knowledge, and practical exposure necessary to excel in their careers.

Mission: To bridge the gap between academia and industry by equipping learners with hands-on expertise, making them employable and future-ready.

CHAPTER 3: INTERNSHIP PART

The internship program was structured into weekly modules, each focusing on key areas of Java enterprise development, Spring Boot microservices, and Python-based NLP machine learning. The program combined guided training with hands-on practice, ensuring a balance of conceptual clarity and real-world application.

- Initial Phase: The internship started with Java 17 fundamentals, Spring Boot framework concepts, and REST API design. We learned to implement user registration/authentication APIs and connect to a MySQL database using Spring Data JPA.
- Implementation Phase: Focused on microservices infrastructure. We built and configured the service-registry (Eureka Server) and the api-gateway (Spring Cloud Gateway) to route client-side AJAX requests seamlessly to downstream services.
- Evaluation and NLP Phase: We created the python-nlp-service using Flask. We loaded the pre-trained GPT-2 model and implemented mathematical perplexity score algorithms based on token cross entropy loss, alongside text burstiness metrics.
- Final Stage: Completed the integration with the frontend-service using Thymeleaf templates. We debugged CORS issues, resolved port mismatch errors between frontend and gateway, calibrated threshold logic, and successfully compiled all microservices.

ACTIVITY LOG FOR THE 1 WEEK

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Person In charge Signature
Day-1 (05-05-2025)	Study of Java 17 features, Maven dependencies, and Spring Boot starter framework.	Gained conceptual clarity on Spring dependency injection and Maven child module configurations.	
Day-2 (06-05-2025)	Designing REST API architecture and database mapping for user-service endpoints.	Understood HTTP method structures and RESTful design patterns for enterprise backends.	
Day-3 (07-05-2025)	Creating JPA entity mappings (User.java and Contact.java) and connecting to local MySQL database.	Learned ORM entity-relationship configurations and database connectivity settings.	
Day-4 (08-05-2025)	Implementing UserRepository and ContactRepository using Spring Data JPA.	Acquired skills in writing Spring Data interfaces and standard database query methods.	
Day-5 (09-05-2025)	Writing UserService logic for registration, authentication, and contact submissions.	Learned backend validation logic and encryption guidelines for storing credentials.	
Day-6 (10-05-2025)	Developing UserController endpoints and running API validation testing via Postman.	Understood how to test REST controllers and inspect JSON payloads.	

WEEKLY REPORT

WEEK – 1 (From Dt 05-05-2025 to Dt 10-05-2025)

Objective of the Activity Done: Spring Boot REST API Development - JPA, Hibernate, MySQL integration, and User Service implementation.

Detailed Report: The first week of the internship was dedicated to building a foundation in Spring Boot REST API development. We learned to configure Maven multi-module dependency layers, create JPA entity mappings, and connect a relational database (MySQL). By implementing user authentication and contact-saving logic in the User Service layer, we gained hands-on experience in business validation. This week culminated in testing all HTTP POST/GET endpoints using Postman, ensuring robust JSON mappings.

ACTIVITY LOG FOR THE 2 WEEK

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Person In charge Signature
Day-1 (12-05-2025)	Introduction to microservice architecture, service discovery, and gateway patterns.	Understood the benefits of microservices over monolithic backend designs.	
Day-2 (13-05-2025)	Setting up and enabling Spring Cloud Netflix Eureka Server in service-registry.	Learned how to configure a centralized Service Discovery Server.	
Day-3 (14-05-2025)	Integrating Eureka Clients into existing Spring Boot microservices.	Gained skills in registering services dynamically with the Eureka Server.	
Day-4 (15-05-2025)	Creating the API Gateway module using Spring Cloud Gateway.	Understood the role of a centralized entry gateway for API routing.	
Day-5 (16-05-2025)	Configuring path-based routing rules in gateway's application.yml.	Learned how to map route predicates and downstream load balancer (lb://) targets.	
Day-6 (17-05-2025)	Verifying client registration status on the Eureka Dashboard.	Successfully mapped and verified dynamic route forwarding.	

WEEKLY REPORT

WEEK – 2 (From Dt 12-05-2025 to Dt 17-05-2025)

Objective of the Activity Done: Microservices Architecture Setup - Eureka Service Discovery Registry and Spring Cloud API Gateway routing.

Detailed Report: The second week focused on setting up the microservice ecosystem. We built the centralized Service Discovery registry (Eureka) and registered our microservices. To consolidate all public requests, we implemented the API Gateway. We configured routing rules so that `/api/user/\*\*` requests are automatically routed to the User Service, and other routes forward to their respective services, creating a unified API infrastructure.

ACTIVITY LOG FOR THE 3 WEEK

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Person In charge Signature
Day-1 (19-05-2025)	Setting up Python virtual environments and pip installing PyTorch, Transformers, and Flask.	Gained skills in setting up machine learning python workspaces.	
Day-2 (20-05-2025)	Loading pre-trained GPT-2 model and tokenizer from Hugging Face repository.	Understood how transformer models process text tokens in AI pipelines.	
Day-3 (21-05-2025)	Implementing text tokenization and vocabulary mapping using GPT2Tokenizer.	Learned how input text is converted to token tensors for modeling.	
Day-4 (22-05-2025)	Exploring the mathematical concepts of perplexity and burstiness metrics.	Acquired foundation in metrics that distinguish human vs AI styles.	
Day-5 (23-05-2025)	Developing the perplexity calculation logic based on GPT-2 token cross-entropy loss.	Learned how to evaluate sentence prediction probability.	
Day-6 (24-05-2025)	Building Flask endpoints to accept POST requests and return NLP evaluation metrics.	Created a functional Python Flask REST API.	

WEEKLY REPORT

WEEK – 3 (From Dt 19-05-2025 to Dt 24-05-2025)

Objective of the Activity Done: Python NLP Service Development - PyTorch, Hugging Face GPT-2 Integration, Perplexity, and Flask REST API.

Detailed Report: The third week transitioned into the AI and Natural Language Processing (NLP) layer. We developed the Python NLP backend using Flask. We loaded the pre-trained Hugging Face GPT-2 model and engineered the core text detection logic. We implemented mathematical perplexity evaluation (which uses cross-entropy loss of predicted next tokens) and word burstiness metrics, exposing these functions via a unified `/detect` endpoint.

ACTIVITY LOG FOR THE 4 WEEK

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Person In charge Signature
Day-1 (26-05-2025)	Creating the Java detection-service module to act as a bridge.	Understood how to link Spring Boot with Python backends.	
Day-2 (27-05-2025)	Configuring Spring WebClient bean in detection-service for HTTP requests.	Learned reactive client configurations in Spring Boot.	
Day-3 (28-05-2025)	Implementing DetectionController to receive text analysis payloads.	Gained skills in routing requests dynamically between backends.	
Day-4 (29-05-2025)	Registering detection-service as a client in the Eureka registry.	Ensured the gateway could discover the new detection microservice.	
Day-5 (30-05-2025)	Implementing robust exception handling for connection timeouts.	Acquired skills in writing resilient backend applications.	
Day-6 (31-05-2025)	Conducting integration tests from Gateway to python-nlp-service.	Verified correct data parsing and HTTP response mapping.	

WEEKLY REPORT

WEEK – 4 (From Dt 26-05-2025 to Dt 31-05-2025)

Objective of the Activity Done: Java-Python Integration - Spring WebClient proxy, REST communications, and microservice integration.
Detailed Report: The fourth week was dedicated to integrating our Spring Boot backend with the Python NLP service. We created a separate microservice (`detection-service`) that registers with Eureka. Using Spring's reactive `WebClient`, we established an inter-service HTTP connection to proxy text analysis payloads to the Python Flask server, parse the returns, and send formatted replies back to the gateway.

ACTIVITY LOG FOR THE 5 WEEK

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Person In charge Signature
Day-1 (02-06-2025)	Creating the frontend-service module with Thymeleaf dependencies.	Learned to build backend-for-frontend Spring projects.	
Day-2 (03-06-2025)	Designing login, registration, and contact templates with HTML/CSS.	Acquired skills in standard HTML structure and CSS styling.	
Day-3 (04-06-2025)	Integrating Bootstrap framework and setting up navigation headers.	Gained skills in responsive web design templates.	
Day-4 (05-06-2025)	Writing FrontendController mappings for pages and form controllers.	Understood MVC controller routing and model attributes.	
Day-5 (06-06-2025)	Connecting registration, login, and contact forms to the API Gateway.	Learned to submit form data and invoke load-balanced RestTemplate.	
Day-6 (07-06-2025)	Implementing HTTP Session management for user authentication.	Acquired skills in tracking logged-in states on the server side.	

WEEKLY REPORT

WEEK – 5 (From Dt 02-06-2025 to Dt 07-06-2025)

Objective of the Activity Done: Frontend Interface Development - Thymeleaf Templates, Bootstrap integration, and user session management.
Detailed Report: The fifth week was dedicated to building the user interface. We created the `frontend-service` module using Thymeleaf templates. We designed responsive web pages for index, login, profile, and contact. We connected these templates to our controllers, which communicate through the API Gateway using a load-balanced RestTemplate. We also implemented HTTP session-based authentication.

ACTIVITY LOG FOR THE 6 WEEK

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Person In charge Signature
Day-1 (09-06-2025)	Designing the core AI text detector page UI.	Gained skills in building analytical web dashboards.	
Day-2 (10-06-2025)	Integrating Plotly.js to plot word frequency bar charts dynamically.	Learned to render interactive JavaScript charts from backend JSON returns.	
Day-3 (11-06-2025)	Debugging the client-side 404 AJAX request routing issue.	Identified that relative URLs must resolve through the API Gateway port (8080).	
Day-4 (12-06-2025)	Correcting the perplexity calculation off-by-one logit alignment bug.	Resolved a critical logic error in the PyTorch cross-entropy inputs.	
Day-5 (13-06-2025)	Calibrating perplexity and burstiness classification thresholds.	Tuned the classifier thresholds to Perplexity 25 and Burstiness 0.20.	
Day-6 (14-06-2025)	Packaging all microservices and conducting final integration tests.	Successfully compiled, package, and ran the integrated project.	

WEEKLY REPORT

WEEK – 6 (From Dt 09-06-2025 to Dt 14-06-2025)

Objective of the Activity Done: Final Integration & Debugging - AJAX resolving, NLP logit shifting, threshold calibration, and final deployment.

Detailed Report: In the final week, we integrated the core detector page. We utilized JavaScript to issue AJAX fetch calls and Plotly.js to render word frequency charts. We resolved two major bugs: a routing port mismatch that was causing 404 connection errors in the browser, and an NLP logit alignment bug in Python. We calibrated the perplexity threshold to 25 and burstiness to 0.20, achieving successful detection of human vs AI text.

CHAPTER 4: OUTCOMES DESCRIPTION

Describe the work environment you have experienced (in terms of people interactions, facilities available and maintenance, clarity of job roles, protocols, procedures, processes, discipline, time management, harmonious relationships, socialization, mutual support and teamwork, motivation, space and ventilation, etc.)

The work environment at SkillDzire was structured, learner-centric, and highly supportive. My job role was clearly defined, with well-communicated objectives and expectations that ensured clarity throughout the internship. The organization maintained a strong sense of discipline, professionalism, and time management, which helped interns stay focused and productive.

Interactions with mentors and peers were collaborative and encouraging. The team fostered mutual support, teamwork, and knowledge sharing, creating a harmonious environment where interns could learn, clarify doubts, and exchange ideas freely. The company's human-centric approach and dedication to bridging industry skill gaps were evident in the way tasks and projects were organized.

Overall, the environment at SkillDzire combined discipline, collaboration, and innovation, enabling me to learn effectively, stay motivated, and experience a professional culture that emphasized both individual growth and teamwork.

Describe the real time technical skills you have acquired (in terms of the job-related skills and hands on experience)

During the internship at SkillDzire, I gained valuable hands-on experience by working on real-time datasets and projects that simulated industry scenarios. The key technical skills I acquired include:
• Microservices Architecture: Designed and integrated a 5-module microservices network utilizing Spring Cloud Netflix Eureka for service registry and Spring Cloud Gateway for centralized API routing.
• Java Spring Boot: Acquired professional experience in REST API development, JPA entity mapping, Hibernate ORM configurations, and load-balanced RestTemplate API calls.
• Python Flask & PyTorch: Learned to load pre-trained Hugging Face transformer models (GPT-2) and implement analytical REST APIs using Flask.
• Natural Language Processing (NLP): Mastered core NLP concept calculations including perplexity (cross-entropy probability) and burstiness (word frequency distribution) to identify machine patterns.
• Frontend Technologies: Gained proficiency in Thymeleaf templating, dynamic form submissions, user session management, and Plotly.js chart visualization.
• Debugging & Integration Testing: Developed systematic debugging skills, resolving network port mismatches, CORS errors, and mathematical token-shifting issues.

Describe the managerial skills you have acquired (in terms of planning, leadership, team work, behavior, workmanship, productive use of time, weekly improvement in competencies, goal setting, decision making, performance analysis, etc.)

The internship helped me acquire several key managerial skills:
• Planning & Goal Setting: I learned to break larger tasks into smaller milestones, define weekly objectives, and set achievable goals, which improved project flow and organization.
• Time Management: By prioritizing tasks and following structured schedules, I was able to meet deadlines effectively while maintaining consistency in daily work.
• Workmanship & Performance Analysis: I developed a habit of reviewing my work for accuracy and quality, analyzing outcomes, and making improvements for better results.
• Decision-Making: I gained confidence in evaluating multiple approaches, weighing pros and cons, and choosing the most effective solutions during technical problem-solving.
• Teamwork & Leadership: I actively contributed to group discussions, collaborated with peers, and provided constructive feedback, while also taking initiative when leadership was required.

Describe how you could improve your communication skills (in terms of improvement in oral communication, written communication, conversational abilities, confidence levels while communicating, anxiety management, understanding others, getting understood by others, extempore speech, ability to articulate the key points, closing the conversation, maintaining niceties and protocols, greeting, thanking and appreciating others, etc.,)

My communication skills improved significantly during the internship through a combination of daily interactions, team activities, and structured reporting. Regular check-ins and weekly report submissions enhanced both my oral and written communication, allowing me to express ideas more clearly and professionally. Participating in team discussions and presenting my progress helped me articulate key points effectively, manage anxiety, and build confidence while speaking. Active listening to mentors and peers not only strengthened my understanding of their perspectives but also ensured that my own inputs were well-received. Additionally, I became more mindful of professional etiquette by maintaining conversational protocols, greeting and thanking appropriately, and appreciating contributions from others. Overall, the internship provided me with opportunities to refine my ability to communicate with clarity, confidence, and professionalism.

Describe how could you could enhance your abilities in group discussions, participation in teams, contribution as a team member, leading a team/activity.

During the internship, I was able to enhance my abilities in group discussions and teamwork through a collaborative and solution-oriented approach. Actively participating in meetings allowed me to share ideas, listen to others' perspectives, and contribute meaningfully to the project's design and implementation. As a team member, I took responsibility for my assigned tasks, ensured timely delivery, and extended support to colleagues whenever needed, which fostered a spirit of mutual growth. These experiences not only strengthened my skills as a dependable contributor but also built confidence in my ability to take initiative and lead activities when required. Overall, the internship prepared me to function effectively as both a supportive team player and a capable leader in professional environments.

Describe the technological developments you have observed and relevant to the subject area of training (focus on digital technologies relevant to your job role)

• **Microservice Orchestration:** Observed standard enterprise patterns involving service registries (Eureka) and API Gateways (Spring Cloud Gateway) to simplify and secure communication in distributed environments.

• **Pre-trained Transformer Models:** Witnessed the rapid growth of large language models. Integrating a pre-trained GPT-2 model locally demonstrated how NLP engines can perform real-time probability estimations.

• **Reactive REST Communication:** Experienced the transition from blocking RestTemplate calls to non-blocking Spring WebClient APIs for high-performance service proxying.

• **Modern Visualization Libraries:** Observed how interactive client-side libraries (like Plotly.js) are used to represent complex data shapes, turning raw token statistics into scannable charts.

• **Cloud & Scalable Solutions:** Working with Python-based ML frameworks and TensorFlow/Keras provided insights into technologies that can be easily deployed on cloud platforms for scalability and efficiency.

CHAPTER 5: RECOMMENDATIONS AND CONCLUSIONS OF THE INTERNSHIP REPORT

The successful completion of this internship demonstrates how an intensive, hands-on learning program can effectively bridge the gap between academic knowledge and industry expectations. My experience at SkillDzire provided me with practical exposure to data analysis, web scraping, machine learning, natural language processing, and deep learning, which are highly relevant in today's data-driven industry.

A key recommendation for future students would be to pursue internships that emphasize real-time projects and practical applications, as these offer the most valuable learning experiences. The opportunity to work on structured and unstructured datasets, build predictive models, and apply modern tools not only enhanced my technical skills but also improved my problem-solving, teamwork, and professional competencies.

In conclusion, this internship was a transformative journey that strengthened both my technical expertise and professional outlook. The methodologies, tools, and industry-aligned practices I gained at SkillDzire will serve as a strong foundation for my future career in data science and artificial intelligence.

Student Self Evaluation of the Short-Term Internship

Student Name : Chenna Reddy Gari Vishnu Vardhan Reddy

Registration No : 229X1A3231

Term of Internship : From: 05/05/2025 To: 20/06/2025

Organization Name & Address : SkillDzire, Hyderabad, Telangana, India.

Name & Address of the Supervisor : SkillDzire Mentor Panel

Rating Scale: 1 is lowest and 5 is highest rank

S.NO	Evaluation Area	1	2	3	4	5
1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Initiative	1	2	3	4	5
4	Interaction with staff	1	2	3	4	5
5	Attitude	1	2	3	4	5
6	Dependability	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Planning and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work	1	2	3	4	5
12	Productivity	1	2	3	4	5
13	Progress of learning	1	2	3	4	5
14	Adaptability to organization's culture/policies	1	2	3	4	5
15	OVERALL PERFORMANCE	1	2	3	4	5

Evaluation by the Supervisor of the Intern Organization

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Term of Internship : From: 05/05/2025 To: 20/06/2025

Organization Name & Address : SkillDzire, Hyderabad, Telangana, India.

Name & Address of the Supervisor : SkillDzire Mentor Panel

Rating Scale: 1 is lowest and 5 is highest rank

S.NO	Evaluation Area	1	2	3	4	5
1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Initiative	1	2	3	4	5
4	Interaction with staff	1	2	3	4	5
5	Attitude	1	2	3	4	5
6	Dependability	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Planning and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work	1	2	3	4	5
12	Productivity	1	2	3	4	5
13	Progress of learning	1	2	3	4	5
14	Adaptability to organization's culture/policies	1	2	3	4	5
15	OVERALL PERFORMANCE	1	2	3	4	5

PROJECT IMPLEMENTATION

Python Code (app.py):

```
from flask import Flask, request, jsonify
from transformers import GPT2Tokenizer, GPT2LMHeadModel
import torch
import nltk
from nltk.probability import FreqDist

tokenizer = GPT2Tokenizer.from_pretrained('gpt2')
model = GPT2LMHeadModel.from_pretrained('gpt2')
app = Flask(__name__)

def calculate_perplexity(text):
    encoded_input = tokenizer.encode(text, add_special_tokens=False, return_tensors='pt')
    if encoded_input.numel() == 0: return 0
    input_ids = encoded_input[0].unsqueeze(0)
    if input_ids.size(1) > 1024: input_ids = input_ids[:, -1024:]
    if input_ids.size(1) <= 1: return 0
    with torch.no_grad():
        outputs = model(input_ids, labels=input_ids)
        loss = outputs.loss
    perplexity = torch.exp(loss)
    return perplexity.item()

def calculate_burstiness(text):
    tokens = nltk.word_tokenize(text.lower())
    if not tokens: return 0
    word_freq = FreqDist(tokens)
    repeated_count = sum(count > 1 for count in word_freq.values())
    return repeated_count / len(word_freq)
```

Flask API Routing implementation (app.py continued):

```
@app.route('/detect', methods=['POST'])
def detect():
    data = request.get_json()
    text = data.get('text', '')
    if not text: return jsonify({"error": "No text provided"}), 400
    perplexity = calculate_perplexity(text)
    burstiness_score = calculate_burstiness(text)

    # Classifier threshold logic (calibrated values)
    if perplexity <= 25 and burstiness_score <= 0.20:
        result = "Likely AI Generated Content"
        status = "error"
    elif perplexity > 25 and burstiness_score > 0.20:
        result = "Likely Human Written Content"
        status = "success"
    elif perplexity > 25 and burstiness_score <= 0.20:
        result = "Possibly AI Generated Content"
        status = "warning"
    else:
        result = "Likely Mixed AI and Human Content"
        status = "info"

    return jsonify({
        "perplexity": perplexity,
        "burstiness_score": burstiness_score,
        "result": result,
        "status": status
    })

if __name__ == '__main__':
    app.run(port=5001, debug=True)
```

Java API Gateway Client Proxy Controller Code:

```
package com.detection.detectionservice.controller;

import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.reactive.function.client.WebClient;
import java.util.Map;

@RestController
@RequestMapping("/api/detection")
@CrossOrigin(origins = "*")
public class DetectionController {
    private final WebClient webClient;

    public DetectionController(WebClient.Builder webClientBuilder) {
        this.webClient = webClientBuilder.baseUrl("http://127.0.0.1:5001").build();
    }

    @PostMapping("/detect")
    public ResponseEntity<String> detect(@RequestBody Map<String, String> request) {
        try {
            return webClient.post()
                .uri("/detect")
                .bodyValue(request)
                .retrieve()
                .toEntity(String.class)
                .block();
        } catch (Exception e) {
            return ResponseEntity.status(500).body("{\"error\": \"" + e.getMessage() + "\"}");
        }
    }
}
```

Spring Cloud Gateway Configuration (application.yml):

```
server:
  port: 8080

spring:
  application:
    name: api-gateway
  cloud:
    gateway:
      discovery:
        locator:
          enabled: true
      routes:
        - id: user-service
          uri: lb://user-service
          predicates:
            - Path=/api/user/**
        - id: detection-service
          uri: lb://detection-service
          predicates:
            - Path=/api/detection/**
        - id: frontend-service
          uri: lb://frontend-service
          predicates:
            - Path=/**

eureka:
  client:
    service-url:
      defaultZone: http://localhost:8761/eureka/
```

PROJECT PLOTS & CHARTS

Plotly.js dynamic bar chart metrics details:

The frontend system incorporates Plotly.js to plot a bar chart representing the 'Top 10 Most Repeated Words' extracted during NLTK tokenization. This allows users to visually inspect frequency patterns in their text input.

- X-Axis: Words (stop words like 'is', 'the', and punctuation are removed in the python service first).
- Y-Axis: Counts (integer frequency of occurrence in the input paragraph).

AI-generated texts tend to have very low word repetition rates (low burstiness) and extremely low perplexity values, while human-written content displays typical linguistic variations, higher perplexity levels, and higher burstiness scores.

SAMPLE API DETECTION OUTPUTS

Sample JSON returns from the Gateway API route (/api/detection/detect):

1. AI-Generated Text Return Sample:

```
{
  "perplexity": 12.576888084411621,
  "burstiness_score": 0.08333333333333333,
  "result": "Likely AI Generated Content",
  "status": "error"
}
```

2. Human-Written Text Return Sample:

```
{
  "perplexity": 33.911634567223122,
  "burstiness_score": 0.23214567891234567,
  "result": "Likely Human Written Content",
  "status": "success"
}
```